



Final Project

2025-Microscopic-Image-Detection
rawdata download:

<https://drive.google.com/drive/folders/1WzpETjC1qfGCXA1JmoGHDv5OaiQX3pcZ?usp=sharing>



2025-Microscopic-Image-Detection

Objective: detect and identify positive animal cells under a microscope.

This project aims to:

- Annotate and classify animal cells as **positive** or **negative** using online tools
- Train AI model or apply a basic image-processing algorithm to automatically detect positive cells.
- Compare manual annotation results with automated detection accuracy.

Tools

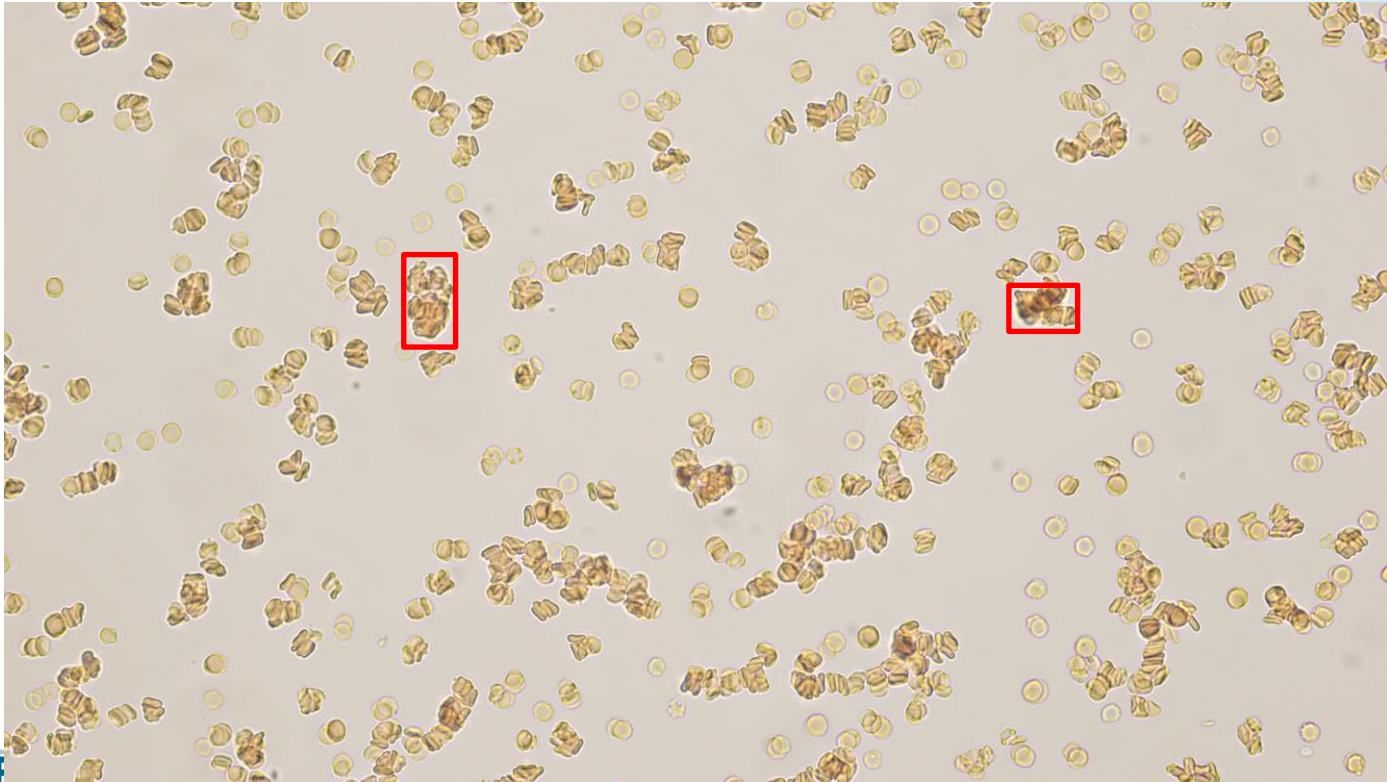
Annotation: **CVAT** , VGG Image Annotator (VIA), Roboflow Annotate,
Analysis: Python (e.g. OpenCV, scikit-image, pytorch)





2025-Microscopic-Image-Detection

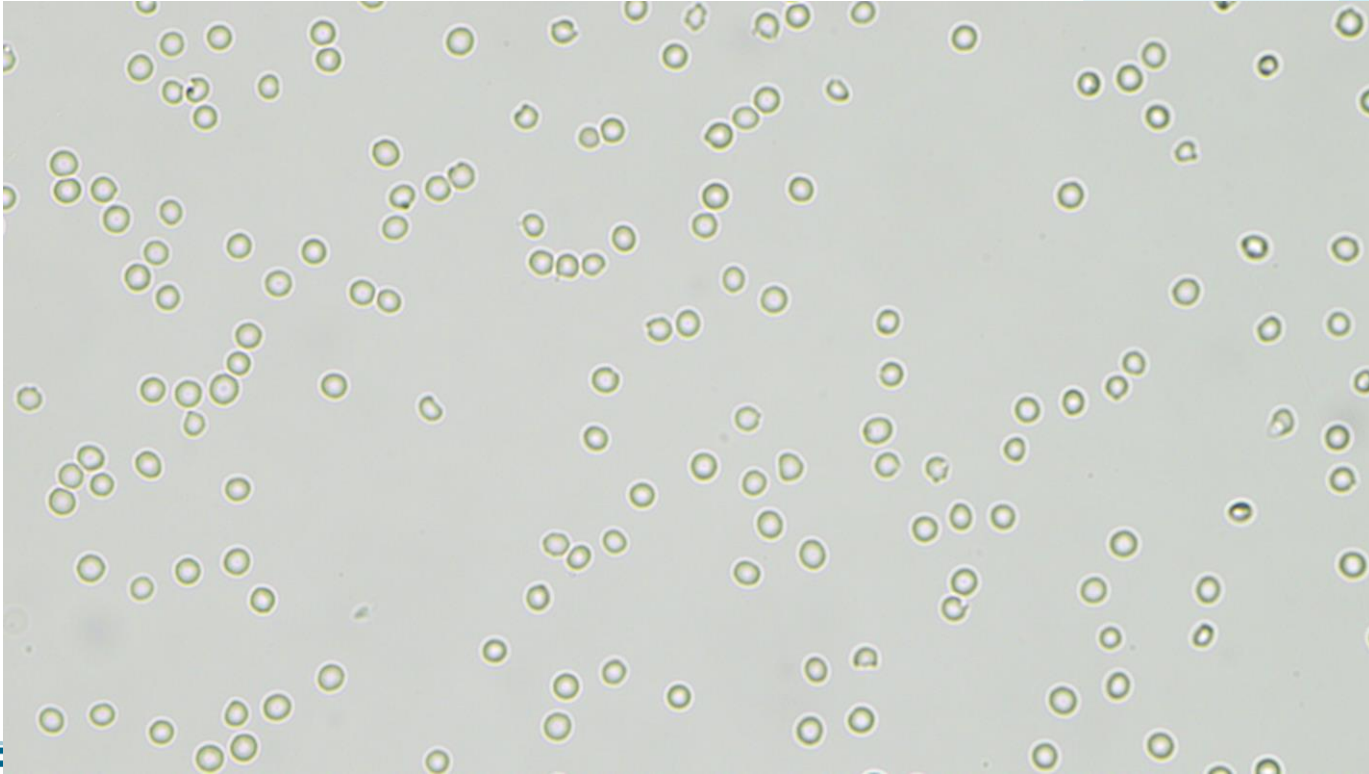
Positive – 需要標註





2025-Microscopic-Image-Detection

Negative - 不需要標註





Annotation

Link: <https://www.cvat.ai/>

CVAT

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Pricing ▾

Labeling Services

Resources ▾

Community ▾

Company ▾

14,691

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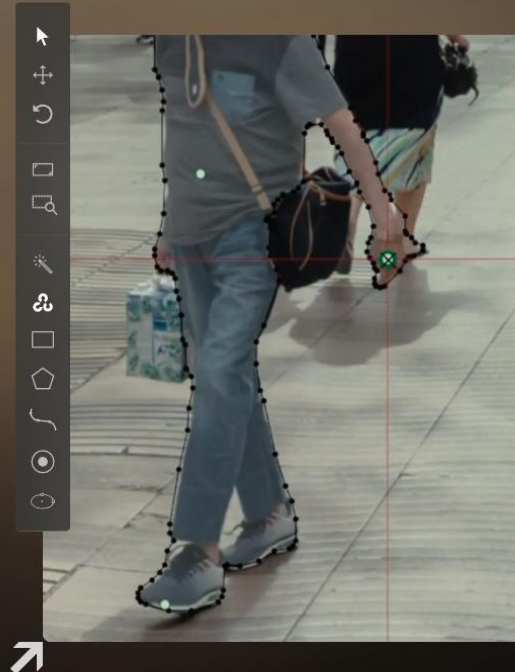
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Leading Data Annotation Platform

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Annotation

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Projects

Tasks

Jobs

Cloud Storages

Requests

Models



ou_chang_yan ▾

Search ...



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No results matched your search...



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Select all

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+ Create a new task

⌚ Create multi tasks

⬇ Create from backup



No results matched your search...



Annotation

Create a new task

Basic configuration

* Name

Test



Project

Select project

Labels

Raw

Constructor

Label name

Any



Add an attribute

Continue

Cancel

* Select files

My computer

Connected file share

Remote sources

Cloud Storage



Click or drag files to this area

You can upload an archive with images, a video, or multiple images

> Advanced configuration



Annotation

Basic configuration

* Name

Test



Project

Select project

Labels



Raw



Constructor

Abnormal



Rect...



Add an attribute +

Continue

Cancel

* Select files

My computer

Connected file share

Remote sources

Cloud Storage



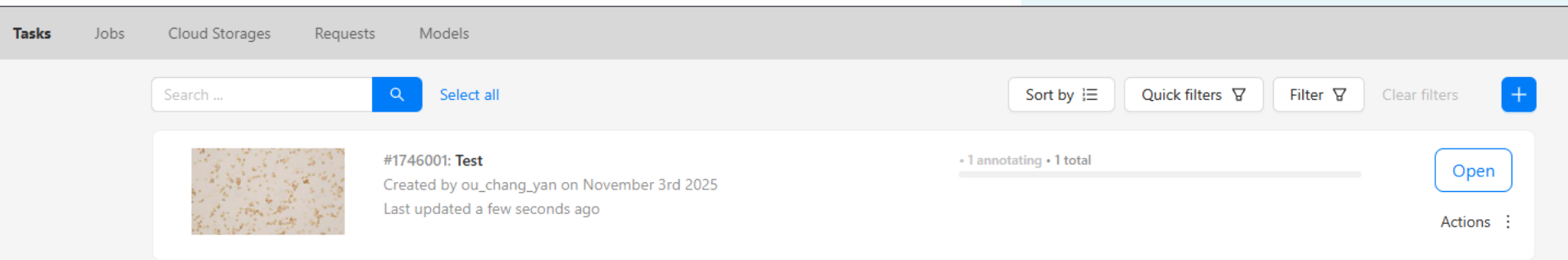
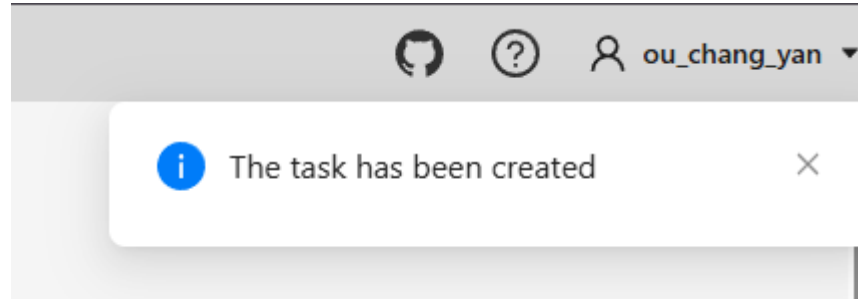
Click or drag files to this area

You can upload an archive with images, a video, or multiple images

> Advanced configuration

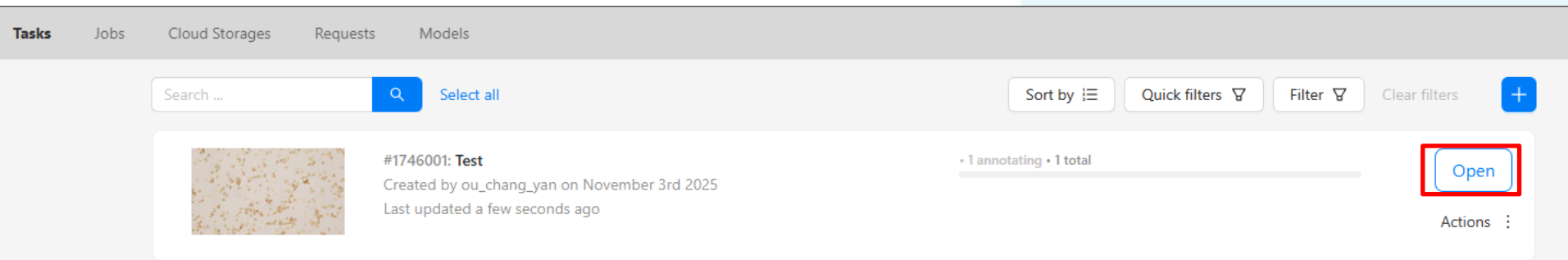
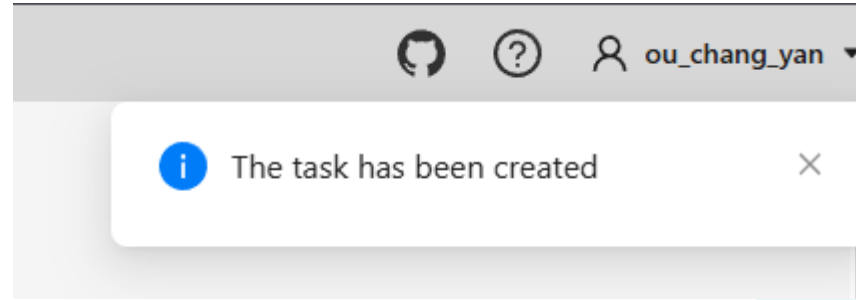


Annotation





Annotation





Annotation

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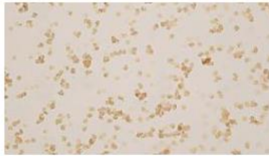
Models



ou_chang_yan

[< Back to tasks](#)

Actions ⋮

Test [✎](#)

Task #1746001 Created by ou_chang_yan on November 3rd 2025

Assigned to

Task description

Issue Tracker [✎](#)[✎](#) Raw[✎](#) ConstructorJobs [Select all](#)

Sort by ≡

Quick filters ▾

Filter ▾

Clear filters

[Job #3200782](#) ⓘ

Created: Nov 3rd 2025 23:49

Updated: Nov 3rd 2025 23:49

Assignee:

Stage:

annotation ▾

State:

new ▾

📅 Duration: a minute

☐ Frame count: 1 (100%)☐ Frame range: 0-0

1



10 / page ▾



Annotation

CVAT Projects Tasks Jobs Cloud Storages Requests Models

Menu Save Undo Redo

杜博台中-20250925-002 (1).jpg 0

Press N to select region and remember to save files(ctrl + s)

ABNORMAL 1 (MANUAL)
124.4x151.0px

Objects Labels Issues

Items: 1 Sort by ID - ascent

1 RECTANGLE SHAPE Abnormal

Appearance

Color by Label Instance Group

Opacity

Selected opacity

☐ Outlined borders ☐ Show bitmap ☐ Show projections



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Search ...

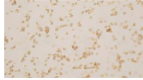
[Select all](#)

Sort by ▮

Quick filters ▾

Filter ▾

Clear filters

#1746001: **Test**

Created by ou_chang_yan on November 3rd 2025

Last updated a minute ago

• 1 annotating • 1 total

[Open](#)

Actions ▮

Upload annotations

Export task dataset

Automatic annotation

Backup Task

Assignee >

View analytics

Quality control

Move to project

Organization >

Delete



1



10 / page ▾



Annotation

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 #1746001: Test
Created by ou_chang_yan
Last updated 3 minutes ago

[Open](#)

Actions :

Export task as a dataset

* Export format

[CVAT for images 1.1](#)

[CamVid 1.0](#)

[Cityscapes 1.0](#)

[COCO 1.0](#)

[COCO Keypoints 1.0](#)

[CVAT for images 1.1](#)

[CVAT for video 1.1](#)

[Datamaro 1.0](#)

[ICDAR Localization 1.0](#)



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[Select all](#)**Export Annotations** [Task #1746001](#)

Started by ou_chang_yan on Nov 3rd 25, 23:58

Expires on Nov 4th 25, 23:58

Finished



100.00%

CVAT for images 1.1

[Download](#)**Create Task**[Task #1746001](#)

Started by ou_chang_yan on Nov 3rd 25, 23:49

Finished on Nov 3rd 25, 23:49

Finished



100.00%



```

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</image>
</annotations>

```



Grade

- 小組書面報告(50%)
- 小組實體報告(40%)
- 小組互評(10%)



Report

包含但不限於:

- 方法與模型選擇
- 流程描述
- 實驗與結果分析
- 技術創新/進階處理方法
- 團隊分工
- 創意發想(e.g. 嚴重程度分級、計數、自動標註.....)
- Bonus: Segmentation



Report

實體報告:

- 請各組上台報告前請將報告的檔案 (.ppt) 上傳至Moodle(一位上傳即可)
- 報告時間請各組必須控制在 15 分鐘內，盡可能將老師提問的時間也加入考量 (Ex. 例如：報告 12 分鐘，留 3 分鐘給老師提問)

書面報告:

- 期末小組書面報告需繳交的檔案為 (.zip)，並請以 “**Group_第幾組**” 格式命名，
- 小組書面報告請盡可能詳細撰寫，請附上**程式碼及說明跟標註過的檔案**(可附上連結)
- 每一組請**組長**負責繳交即可。



Report

互評表:

- 每個人都需要附上小組互評表，下方為範例格式
- 繳交的檔案為 (.pdf)
- 並請以 "學號_姓名.pdf" 格式命名

組員	成績
張三	60
李四	100
謝小翰	100